

Understanding Your Data

Series: ONBRD | **Notebook:** 7 of 10 | **Created:** December 2025 | **Last Updated:** 04/04/2026

Exploring What Dynatrace Discovered

With OneAgent deployed, Dynatrace has automatically discovered your infrastructure, processes, and services. This notebook helps you understand what's been found and how to explore your data.

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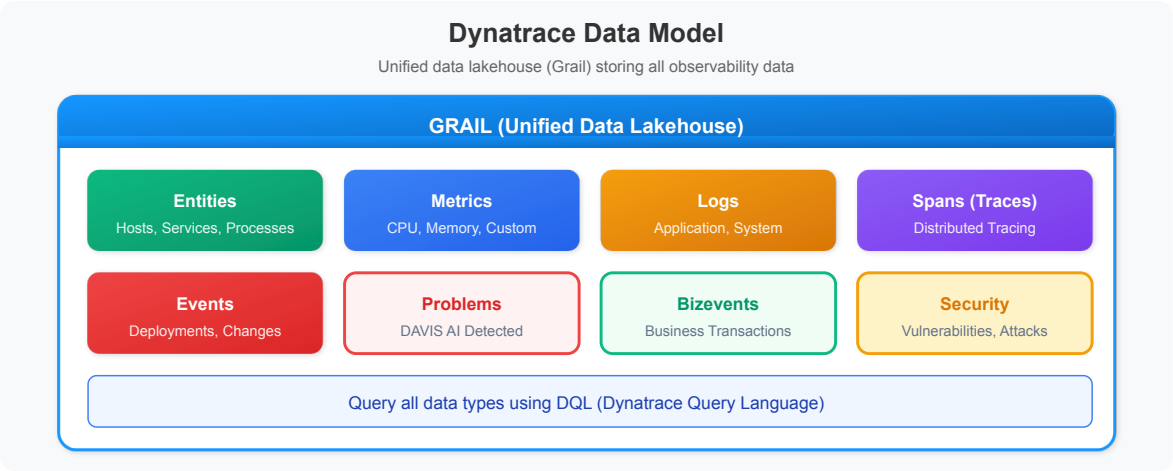
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Prerequisites

- OneAgent deployed on at least one host (ONBRD-05)
- Environment organized with tags (ONBRD-06)
- 15-30 minutes elapsed since deployment for full discovery
- DQL query permissions

1. The Dynatrace Data Model

Dynatrace organizes data into a unified model:

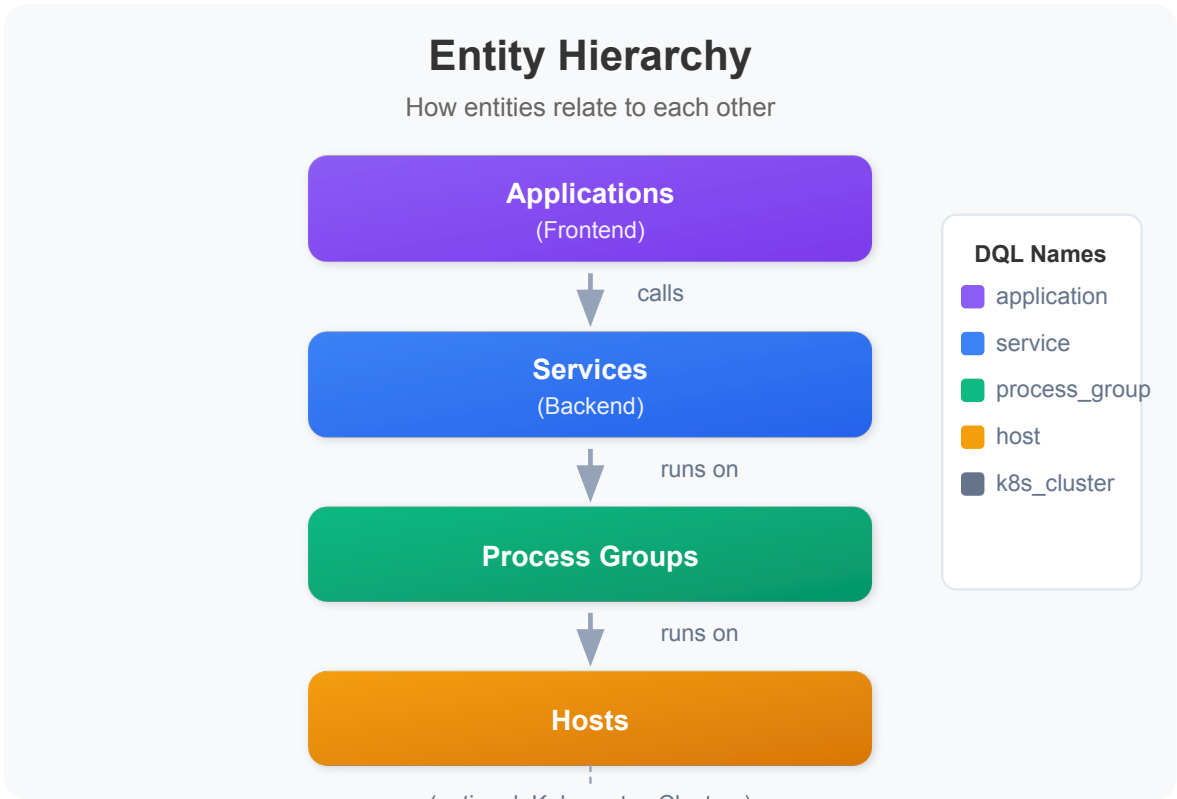


Key Concepts

Concept	Description	Example
Entity	Any monitored component	Host, Service, Process
Metric	Numeric measurement over time	CPU usage, response time
Log	Textual event record	Application log entry
Span	Single operation in a trace	HTTP request, DB query
Event	Point-in-time occurrence	Deployment, config change
Problem	DAVIS-detected issue	Service slowdown

2. Entities and Relationships

Dynatrace automatically discovers and relates entities:



Common Entity Types

Entity Type	DQL Name	What It Represents
Host	<code>dt.entity.host</code>	Physical or virtual machine
Process Group	<code>dt.entity.process_group</code>	Set of identical processes
Service	<code>dt.entity.service</code>	Logical backend service
Application	<code>dt.entity.application</code>	Frontend application
K8s Cluster	<code>dt.entity.kubernetes_cluster</code>	Kubernetes cluster
K8s Namespace	<code>dt.entity.cloud_application_namespace</code>	K8s namespace
K8s Workload	<code>dt.entity.cloud_application</code>	Deployment, DaemonSet, etc.

3. Exploring Topology

The topology view shows the visual representation of your environment.

Location: Infrastructure app → Hosts → Select any host → Dependencies

Understanding Topology Layers

Layer	Shows	Use Case
Applications	Frontend apps, user sessions	User experience
Services	Backend services, APIs	Service dependencies
Processes	Running processes	Process mapping
Hosts	Servers, VMs, containers	Infrastructure view

Reading Topology Views

- **Nodes** = Entities
- **Lines** = Relationships (calls, runs on)
- **Line thickness** = Traffic volume
- **Colors** = Health status (green/yellow/red)

Navigation Tips

- Click any entity to see details
- Use filters to focus on specific services
- Hover over connections to see traffic metrics
- Use the Services app for service-to-service dependencies

4. Data Types in Grail

Grail stores different data types with different retention and query patterns.

Data Type	DQL Fetch	Typical Use
Logs	<code>fetch logs</code>	Troubleshooting, audit
Spans	<code>fetch spans</code>	Distributed tracing
Metrics	<code>timeseries</code>	Performance monitoring
Events	<code>fetch events</code>	Change tracking
Problems	<code>fetch dt.davis.problems</code>	Incident management
Entities	<code>fetch dt.entity.*</code>	Topology queries
Bizevents	<code>fetch bizevents</code>	Business transactions

Data Retention

Default retention varies by data type and license:

Data Type	Typical Retention
Metrics	5 years (aggregated)
Logs	35 days (configurable)
Spans	35 days
Events	35 days
Entity state	Real-time

5. Discovery Queries

Use these queries to understand what Dynatrace has discovered in your environment.

Infrastructure Discovery

```
// Count all entity types in your environment
fetch dt.entity.host | summarize hosts = count()
// Run separately for other types:
// fetch dt.entity.service | summarize services = count()
// fetch dt.entity.process_group | summarize process_groups = count()

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes HOST | summarize hosts = count()
```

```
// List all hosts with key details
fetch dt.entity.host
| fields entity.name,
        state,
        monitoringMode,
        osType,
        cpuCores,
        physicalMemory
| sort entity.name
| limit 100
```

```
// Group hosts by OS type
fetch dt.entity.host
| summarize count = count(), by: {osType}
| sort count desc
```

Service Discovery

```
// List all discovered services
fetch dt.entity.service
| fields entity.name, serviceType
| sort entity.name
| limit 100
```

```
// Group services by type
fetch dt.entity.service
| summarize count = count(), by: {serviceType}
| sort count desc
```

```
// Find services with recent traffic (spans)
fetch spans, from: now() - 1h
| filter span.kind == "server"
| summarize request_count = count(), by: {service.name}
| sort request_count desc
| limit 20
```

Process Discovery

```
// List process groups
fetch dt.entity.process_group
| fields entity.name
| sort entity.name
| limit 50
```

```
// Count process groups
fetch dt.entity.process_group
| summarize count = count()
```

Log Discovery

```
// Check log volume by source
fetch logs, from: now() - 1h
| summarize log_count = count(), by: {log.source}
| sort log_count desc
| limit 20
```

```
// Check log volume by severity
fetch logs, from: now() - 1h
| summarize log_count = count(), by: {loglevel}
| sort log_count desc
```

```
// Sample recent logs
fetch logs, from: now() - 15m
| fields timestamp, loglevel, log.source, content
| sort timestamp desc
| limit 25
```

Kubernetes Discovery (if applicable)

```
// List Kubernetes clusters
fetch dt.entity.kubernetes_cluster
| fields entity.name
| sort entity.name

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes K8S_CLUSTER
// | fields name
// | sort name
```

```
// List namespaces
fetch dt.entity.cloud_application_namespace
| fields entity.name
| sort entity.name
| limit 50

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes K8S_NAMESPACE
// | fields name
// | sort name
// | limit 50
```

```
// List workloads (deployments, etc.)
fetch dt.entity.cloud_application
| fields entity.name
| sort entity.name
| limit 50

// Alternative: Smartscape on Grail (entity.name → name)
// smartscapeNodes K8S_DEPLOYMENT
// | fields name
// | sort name
// | limit 50
```


Problems Discovery

```
// Check for recent problems
fetch dt.davis.problems, from: now() - 7d
| fields timestamp, display_id, title, status, affected_entity_types
| sort timestamp desc
| limit 20
```

```
// Problem summary by status
fetch dt.davis.problems, from: now() - 30d
| summarize problem_count = count(), by: {status}
| sort problem_count desc
```

6. Next Steps

Now that you understand your data:

1. **ONBRD-08: Your First Queries** - Learn DQL fundamentals
2. Explore topology for dependency visualization
3. Review any detected problems
4. Plan which additional hosts to instrument

Discovery Checklist

- ☐ Hosts discovered and showing data
- ☐ Services detected and mapped
- ☐ Process groups visible
- ☐ Topology showing relationships
- ☐ Log ingestion working (if applicable)
- ☐ Kubernetes entities visible (if applicable)

Summary

In this notebook, you learned:

- The Dynatrace data model (entities, metrics, logs, spans)
 - How entities relate to each other
 - How to navigate topology views
 - Different data types stored in Grail
 - Discovery queries for infrastructure, services, and logs
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References

- [Entities](#)
 - [Topology and Dependencies](#)
 - [Grail Data Model](#)
 - [Entity Types Reference](#)
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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.